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CHANGE IN TREE GROWTH BEFORE AND AFTER REWETTING OF DRAINED PEATLANDS



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ABSTRACT

Peatlands are important ecosystems that have been extensively drained for forestry, increasing CO_2 emissions and altering ecosystem functions. Rewetting is now being used as a restoration strategy to reduce emissions and restore peatland ecosystems and biodiversity. However, this could negatively impact tree growth by altering site hydrology and soil conditions. This study investigates changes in tree growth in *Pinus sylvestris* before and after peatland rewetting across eight pairs of peatlands in Sweden in a Before-After-Control-Impact (BACI) design. Tree-ring widths from tree cores collected at varying distances from ditches were measured and converted into area increase and analyzed using linear mixed-effects models. While a negative trend in growth was observed after rewetting and with increasing distance from ditches, the effect could not be detected statistically. The lack of strong effects may be explained by site-specific differences in drainage and rewetting effectiveness and variable water-table changes. Overall, this study indicates that pine tree growth is influenced by peatland drainage and rewetting, but that responses are likely variable and site-specific.

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INTRODUCTION

Peatlands are important ecosystems that play a key role in global carbon storage, hydrological regulation, and in supporting biodiversity. They are characterized by being a type of wetland with a minimum depth of 30 cm peat, which are remains of organic matter, often *Sphagnum* peatmosses, that does not completely decompose due to the water-saturation in the ground which causes anoxic conditions (Rydin, 2013). In Sweden, many peatlands have historically been drained for forestry and agricultural purposes, resulting in major alterations to ecosystem functions, including increased carbon dioxide emissions and biodiversity loss. The lowering of the water table leads to oxidation and decomposition of the peat and thereby leads to carbon dioxide being released into the atmosphere (Naturvårdsverket, 2025). As a consequence, peatland restoration and rewetting have become a relevant management strategy for reducing greenhouse gas emissions and restoring ecosystem functions. In Sweden, 2800ha drained wetlands were rewetted in 2025, an effort which had the effect of limiting CO2 emissions by 12 200 t (Naturvårdsverket, 2025). Pine trees grow on some peatlands and are a flood-tolerant coniferous species, and therefore able to cope with some of the water stress in peatlands. Pine trees grown on peatlands have a limited ring-width growth and are often smaller in these conditions and also show special adaptions to peatland conditions in their xylem (Dinella et al., 2021).

A standard method to quantify tree growth is measuring the width of the tree rings. Tree rings consist of xylem cells, and increased growth results in the production of more cells, leading to wider tree rings. Tree rings are composed of two different parts of xylem, earlywood and latewood. Earlywood tracheid cells are wide and thin walled, making sap transport more efficient in the early growing season. Latewood tracheid cells, made in the late growing season, are narrower and more thick-walled to ensure increased mechanical stability for the tree (Shishov et al., 2023). Thus the “rings” seen in tree rings are latewood, and the spaces found in between are earlywood.

The xylem cells can be affected by the water table levels in peatlands, since root growth as well as mineral and nutrient uptake require oxygen in their processes. A high water table therefore limits the tree as it is an anoxic environment, which becomes even worse in stagnant water (Rydin, 2013). In a greenhouse experiment on pine seedlings, wet conditions resulted in reduced growth and cell formation compared to drier treatments (Edvardsson et al. 2024). The production of xylem cells can furthermore be influenced by temperature, soil moisture, precipitation, and drought. Ongoing climate change is also likely to have an impact on tree growth, as many environmental factors are changing, such as increase in drought and lengthening of the growing season (Shishov et al., 2023).

Specific trees have, however, been shown to have some adaptations to living in waterlogged conditions. One study on oaks in Germany showed that, while many trees died with the rising water table, the ones who lived had made cellular adaptions to survive. They showed a decrease in the hydraulic diameter of the vessels, meaning that the water-transporting

vessels became smaller, but also an increase in vessel number and ring width. The increase in ring width was discussed to be a consequence of the release effect, where more nutrients and sunlight was available due to the death of the other trees (Resente et al., 2024). Another study on alder trees found that while alder benefited from wet conditions, permanent waterlogging had a strong negative effect on tree growth (Anadon-Rosell et al., 2022).

Another factor that can impact tree growth on managed peatlands is distance to ditches. When a peatland is drained, the water level will be lowest near the ditch (Bring et al., 2022), and a study in drained conditions on pine trees (*Pinus sylvestris*) showed that as the distance from ditches decreases, the tree ring width increases. This was reasoned to be because the ditches remove water, which lowers the water table and improves soil aeration and uptake of nutrients (Parasyris, 2024).

The aim in this study is to analyze the change in tree growth for pine trees before and after rewetting peatlands. A further aim is to analyze if the distance from the ditch affects tree growth before and after the rewetting. The following hypothesizes were tested:

- The trees will have a smaller annual area increase after the rewetting, as the growth was impeded when the water table was raised. This creates a more anoxic environment where the trees do not thrive, since many of their growth processes require oxygen.

- The trees closest to the ditches will, in the drained sites, have less anoxic conditions, and show higher growth than ones further away from the ditch. After rewetting, they will show a larger decrease in growth compared to trees further away from the ditch.

METHOD

To study the effect of rewetting on tree growth, eight pairs of peatlands in Sweden were selected and sampled in June 2024: Stormyran, Nybo-stormyran, Långmyrberget, Sör-lappmyran, Mossaträsk, Vändåtberget and Tväråberget (see map). All peatlands had been drained for a minimum of 50 years, and each pair consisted of one site that was recently rewetted and one control site that was still drained, forming a Before-After Control-Impact (BACI) study design. The control sites were included to distinguish the effects of rewetting from natural variation over time.

One rewetted and one drained control site were sampled in each pair of peatlands, except for Mossaträsk, which had one drained and two rewetted sites, and Tväråberget, where only a drained control site was sampled. At each site, tree cores were taken at breast height (1.3 m) from three pine trees (*Pinus sylvestris*) located at varying distances from the ditch. In Långmyrberget, spruce trees (*Picea abies*) were sampled instead. Tree circumference was measured, as well as distance to the ditch, with one tree sampled from each of three distance categories: near (1-5m), mid (7-16m) and far (16-30m). Stormyran, Nybo-stormyran, Långmyrberget, Vändåtberget and the easternmost site at Mossaträsk were rewetted in

2021. Sör-Lappmyran and the westernmost site at Mossaträsk were rewetted in 2014, while Tväråberget was rewetted in 2020. Water-table levels were also recorded at the sites.

The tree cores were collected using an increment borer, a hollow drill that enables the extraction of a cylindrical wood sample. The tree cores were stored in plastic straws before being extracted and glued into grooves in wooden boards and labelled with identification number and orientation marks. To clearly distinguish the annual tree rings, a flat and smooth surface is required. The cores were therefore sanded, first using a belt sander with P80 grit, followed by orbital sanding with P120, P240 and P320 grit sandpaper, and finally polished manually with P2000 grit sandpaper. All tree cores were scanned at a resolution of 2400 dpi using a CanoScan LiDE 400 flatbed scanner (Canon Inc., Tokyo, Japan) together with the VueScan software (Hamrick Software, Phoenix, AZ, USA).

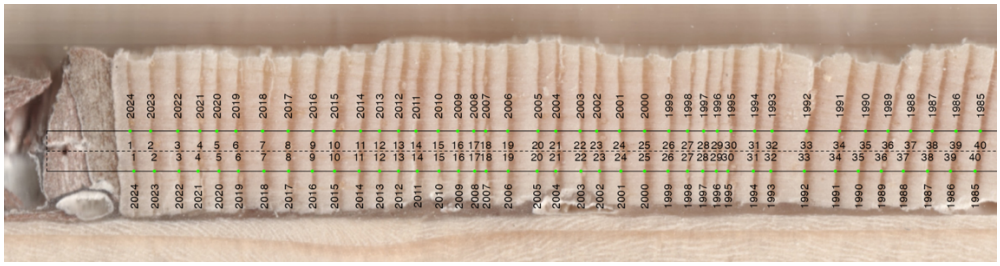


Figure 1: Shows a picture of a tree core being measured in the program MtreeRing using the watershed method with the method for inclined tree rings enabled.

The width of each tree ring in each tree core was measured using the program MtreeRing in R (Shi, Xiang, 2026), as shown in *Figure 1*. The associated Shiny application was modified to enable the use of all available measuring methods (asplundj, 2026). The primary measuring method used was the watershed method, in which inclined tree rings could be measured by defining two reference points for each ring. The software automatically detects the tree rings, however, all samples were carefully inspected and manually corrected due to frequent detection errors. An additional method, measuRing, was used selectively for samples with straighter rings that were hard to detect using the watershed method, as the method measuRing performed better in automatic ring detection. However, the method measuRing was unable to measure inclined tree rings and therefore used quite sparsely.

The ring widths and circumference of the trees were used to calculate the area increase in mm^2/year for each tree core. A bark layer of an estimated 0.5 cm was subtracted from the radius of each tree. The area was calculated to get a more even measurement, because larger and wider trees must produce more cells, as they are distributed across a larger area, to get the same size tree rings as a smaller tree. The logarithmic area was used in the analysis as it showed more evenly distributed residuals.

A linear mixed-effects model (LMM) was used for the analysis, using the R packages lme4 (Bates et al., 2015) and lmerTest (Kuznetsova et al., 2017). Fixed effects included the factors Period (before vs. after rewetting), Treatment (rewetted vs. still drained control sites), and

the continuous variable centered distance from the ditch. Area, Site and Tree were included as random effects to account for the repeated measurement. Each area included two sites: one rewetted site and one drained control site. Here, “Area” refers to the location, “Site” indicates whether a site was rewetted or a drained control, and “Tree” identifies the individual sampled trees within each site. The inclusion of random effects accounts for the non-independence of observations within the dataset.

Some peatlands were rewetted in 2021 and only had two years post rewetting. Preliminary analyses comparing two-year and nine-year before and after rewetting showed only minor differences in the effect of rewetting with overlapping confidence intervals. To ensure evenness and comparability among areas and periods before and after, analyses were restricted to two years before and two years after rewetting. The data from the year of rewetting was excluded from the analysis, as it is unknown in which season the rewetting was done, making it unclear whether it should be classified as occurring before or after rewetting.

The likely year of drainage was examined by looking at historical areal photographs from the mapping service *Min karta* made by Lantmäteriet from 1960 and 1975 and seeing if ditches were present. The graphs were then examined for spikes to see the likely timing of the drainage.

All statistics were done in R (v.4.5.3; R Core Team, 2026) and the graphs were illustrated with the packet ggplot2 (H. Wickham 2016).

RESULT

A total of 42 tree cores from eight different areas were analyzed. One core, collected from the tree closest to the ditch at Sör-lappmyran was cracked, making it difficult to accurately determine the starting year, and was therefore excluded from the analysis. The area increase per year for each tree core can be seen in *Figure 2 and 3*.

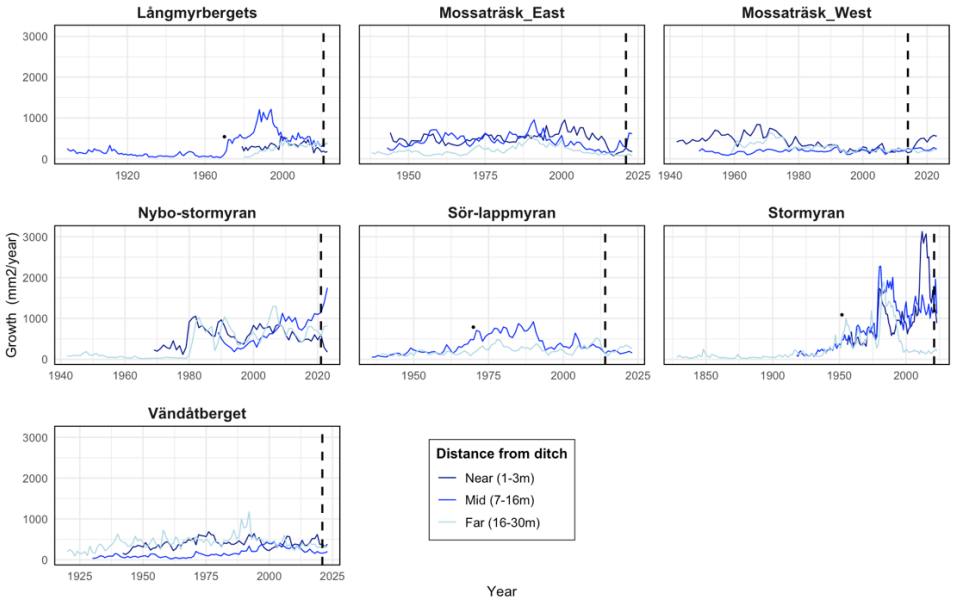


Figure 2: Tree growth in all rewetted sites. Each line shows one tree core from one tree. The color of the lines indicates the trees' distance from the ditch, where dark blue is 1-3m, blue 7-16m and light blue 16-30m away from the ditch. The black points indicate the likely year of drainage, and the dotted line indicates the year of rewetting.

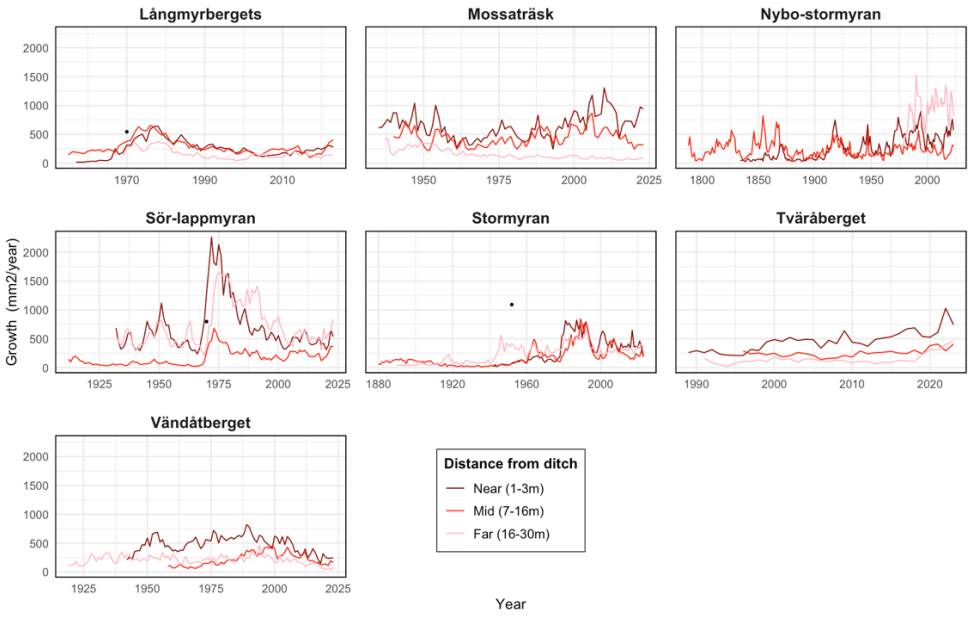


Figure 3: Tree growth in all control sites that are still drained. Each line shows one tree core from one tree. The color of the lines indicates the trees' distance from the ditch, where dark red is 1-3m, red 7-16m and light red 16-30m away from the ditch. The black points indicate the likely year of drainage.

For some sites, the year of drainage could be seen clearly in the graphs as the tree growth increased, for example in the control site for Sör-Lappmyran and rewetted site for Nybo-stormyran (*Figure 2 and 3)*. Based on the hillshade maps of year 1960 and 1975 from Min karta by Lantmäteriet, all peatlands looked to have ditches in 1975, but only Vändåtberget, Stormyran rewetted, Nybo-stormyran control, and possibly Tväråberget, Nybo-stormyran

rewetted and Mossaträsk had ditches in 1960. However, in the graphs, only some sites show a clear peak that indicates response to drainage. For both rewetted and control sites, those included Långmyrberget, Sör-lappmyran and Stormyran.

The impact of time after rewetting was investigated by looking at the difference between a two-year and nine-year window in Sör-lappmyran and westmost Mossaträsk, which were rewetted in 2014. The two-year window was extracted by only analyzing two years before and after the rewetting event. The difference is small and with largely overlapping confidence intervals (*Figure 4*).

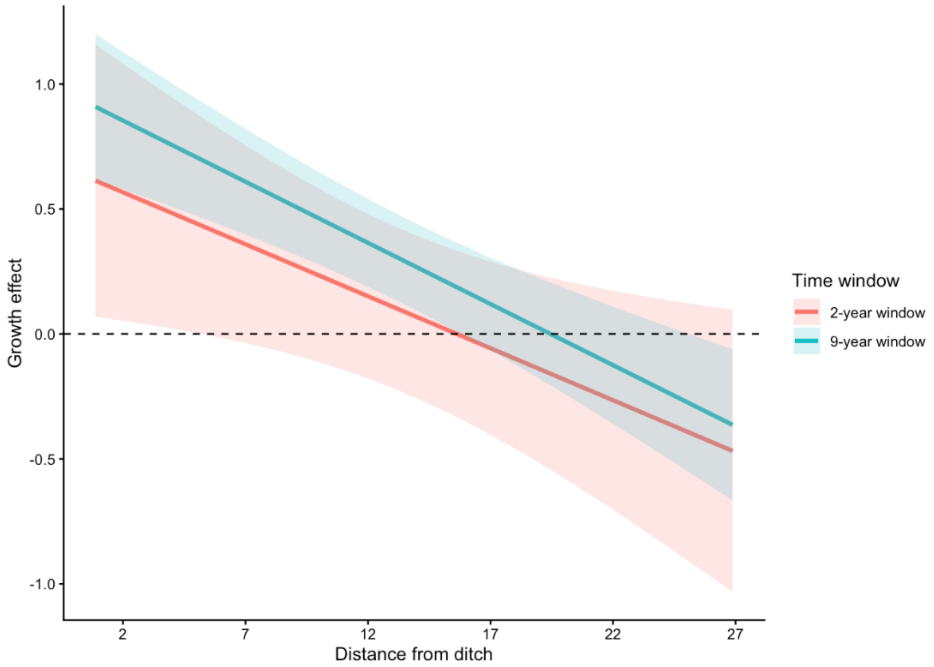


Figure 4: The impact of time since rewetting on growth on two rewetted sites, Sör-lappmyran and the westernmost site at Mossaträsk, in relation to distance from the ditch. The dotted line at 0.00 shows the reference line, calculated from the control sites. The red line shows the effect over a two-year period and the blue line over a nine-year period. The shaded area shows the 95% confidence intervals.

The ANOVA showed no significant interaction between treatment and period ($F_{1,119}$ = 2.82, p = 0.096) or for the three-way interaction between treatment, period and the centered distance to the ditch ($F_{1,119}$ = 0.74, p = 0.391). However, a trend could be observed, where tree growth tended to decrease after rewetting relative to drained control sites (β = −0.145, p = 0.096). A negative trend could also be observed in respect to distance to the ditch, where trees closer to the ditch had more growth than trees further from the ditch, and where rewetted sites generally had less growth than the control sites (Figure 5). However, the confidence intervals are large and show substantial uncertainty.

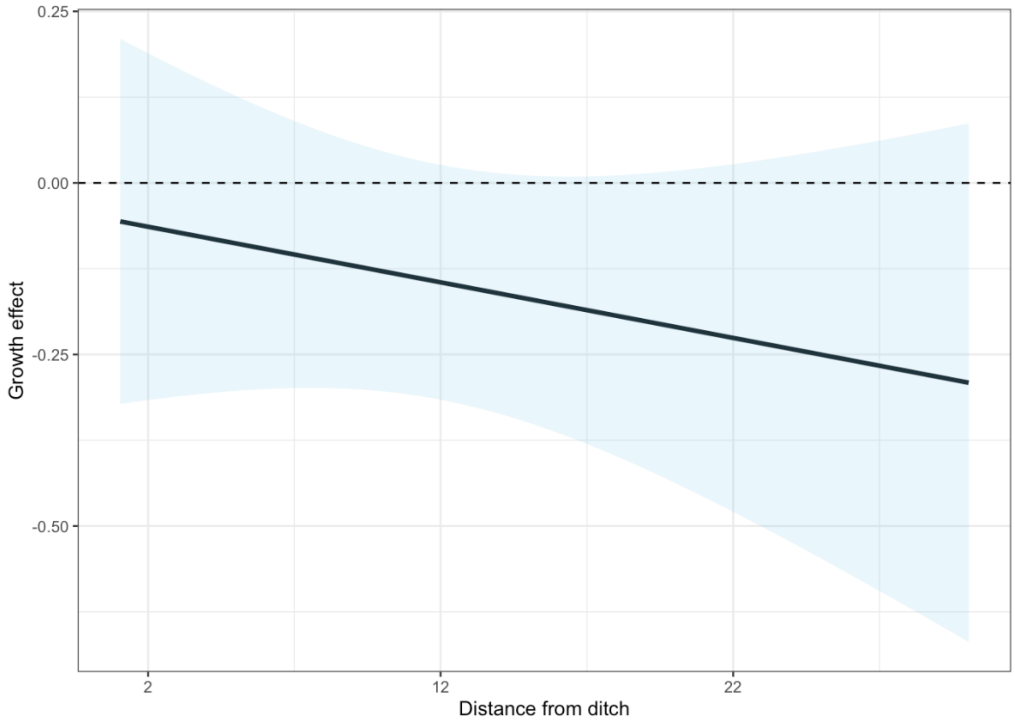


Figure 5: This graph shows the effect of rewetting drained peatlands on tree growth, in relation to distance from ditch. The dotted line at 0.00 shows the reference line, calculated from the control sites. The y-axis shows growth effect in mm^2/year compared to the reference at 0.00 and the x-axis shows distance from ditch in meters. The shaded area shows the 95% confidence intervals.

The analysis was also tested without the one site which sampled spruce instead, but it did not impose a different result. Additionally, the model was also tested using ring width as the response variable, which likewise did not change the outcome.

Table 1: Shows the water table level depth in all of the studied sites except Långmyrberget and Tväråberget. The samples were collected in 2024 together with the tree cores after the rewetting, at both control and rewetted sites. WTD = Water table depth

Site	Range WTD	Mean WTD
Långmyrberget - Control	9–21	13,4
Långmyrberget - Rewetted	8–17	11,5
Mossaträsk - Control	20–52	25
Mossaträsk East - Rewetted	3–28	13
Mossaträsk West - Rewetted	12–79	35
Nybo-stormyran - Control	24–80	57
Nybo-stormyran - Rewetted	7–14	10
Sör-lappmyran - Control	10–68	43
Sör-lappmyran - Rewetted	3–10	7
Stormyran - Control	8–44	25
Stormyran - Rewetted	19–55	24
Vändåtberget - Control	5–55	28
Vändåtberget - Rewetted	8–30	18

The water table depths varied among sites, but the rewetted sites showed lower levels than the control sites, except for the westernmost rewetted Mossaträsk site (*Table 1*). Some sites did however have quite small differences, such as Långmyrberget and Stormyran, whereas the biggest differences could be seen in Nybo-stromyran and Sör-lappmyran (*Table 1*).

Discussion

Although the linear mixed-effects model showed no statistically significant effects, the observed negative trend suggests that rewetting reduces tree growth, and that this effect becomes more pronounced with increasing distance from the ditch. Therefore, the results do not provide support for my first hypothesis that annual growth will decline following rewetting. However, the observed trend is consistent with the first hypothesis and with the expectation that higher water tables create more acidic and anoxic soil conditions which limit processes that require oxygen (Dinella, 2021). In a study with *C. thyoides* trees on peatlands, this was found to be the case (Atkinson et al., 2025).

Likewise, the results do not support the second hypothesis that trees closest to the ditch would experience the strongest reduction in growth. Instead, the observed trend suggests a stronger negative response at greater distances from the ditch. This may initially appear counterintuitive, as it was expected that trees closer to the ditch would be more strongly affected due to the presumably lager changes in water table level. As in a meta-study on groundwater levels, it was found that following rewetting, the increase in water table was substantially higher nearest to the ditch and following drainage the opposite, with lowest

water table closest to the ditch (Bring et al. 2022). So, the observed negative trend suggests that the relationship between distance to ditch and hydrological response is more complex than first thought.

In the drained condition, trees closer to the ditch may experience a lower water table than the ones further away, as the water table lowers near the ditch as the water flows away into the ditch. In the rewetted condition, trees near the ditch may experience a larger absolute change in water table (Bring et al., 2022) yet may still maintain comparatively lower water table levels than further away, depending on site hydrology. The trees further away may, with the slight increase in water table, reach a critical point in which their growth response is substantially affected (Anadon-Rosell et al., 2022).

An additional explanation may be that ditch excavation created elevated peat banks near the ditch. This would allow the trees closest to the ditch to maintain a greater portion of their root system above the water table, which could explain why the trees nearest to the ditch were not as affected. Similar patterns have been observed in peatlands where trees growing on slightly elevated hummocks experience comparably dryer conditions and have wider tree rings (Janecka et al., 2025). Furthermore, while temporal effects could be considered, the results in this study do not support this. *Figure 4* only shows slight differences between two-year and nine-year responses. This suggests that time does not play a significant role in the level of growth response a rewetting induces.

Some tree cores only showed minor differences in growth before and after the rewetting, which means that the changes in water table either did not affect the trees or were relatively small. Limited change in water table levels could be because the peatlands were not effectively drained prior to rewetting, or alternatively that the rewetting was not successful and did not manage to raise the water table.

Successful peatland rewetting requires ditch blocking, but also consideration of inflow of water and its connectivity in the landscape. It also requires a high enough water table and returning growth of *Sphagnum* peatmosses (Atkinson et al., 2025). The water table depth, seen in Table 1, shows the differences between the rewetted and control sites, where some, like Sör-lappmyran and Nybo-stormyran shows a large difference, and the rewetted sites have a high water table at most ten centimeters below the surface. Others, for example Stormyran, shows only a small difference in water table, differing only with a few centimeters. This means that all sites may not have been successfully rewetted and the water table levels may have not disturbed the trees root-systems enough that it would cause a response. Under waterlogged conditions, oxygen depletion in the soil can reduce root respiration and metabolism, trigger statal closure, and subsequently reduce photosynthesis and tree growth (Edvardsson et al., 2024). However, the data should be interpreted with a bit of caution, since we are looking at differences across sites and not within the same site before and after rewetting.

As noted above, the lack of a growth response may also be attributed to unsuccessful continuous drainage in some peatlands. The clearest response to drainage was observed at Långmyrberget and Sör-lappmyran, where tree growth increased substantially following drainage (*Figure 2, 3*). This indicates that these sites experienced a pronounced shift in water table level, and that the difference in water table level had an impact on tree growth. In contrast, several other peatlands showed a weaker response to drainage. One possible explanation is that drainage in these sites was not as successful and had a limited effect, potentially due to shallow ditches or ditch construction without consideration or knowledge about the local hydrological conditions. In addition, insufficient maintenance of ditches may have led to gradual overgrowth, reducing water transport out from the peatland and allowing water tables to rise over time. This interpretation is supported by the pattern observed in Långmyrberget and Sör-lappmyran, where the initial post-drainage growth increase appears to decline over time (*Figure 2, 3*), suggesting a gradual reduction in drainage efficiency. This is consistent with findings showing that extensive drainage can initially cause a strong and immediate increase in tree growth, up to three times higher than before, but that effect may diminish over time if drainage is not maintained or if the drainage was small (Linderholm, 1999). It is also likely that the studied sites were not the ones with the most maintained ditches, as the drained peatland forests that are used productively are the ones that have the most properly manage ditches and there may be an incentive to not rewet the most productive ones, as that would mean a bigger economic loss.

Consequently, the recent rewetting measures may not have induced substantial changes in water table levels compared to pre-rewetting conditions, which could explain the lack of clear growth response. Additional growth peaks observed in *Figure 2 and 3*, may instead reflect climatic variability, where for example temperature and precipitation can strongly influence annual tree growth (Dinella, 2021).

Overall, the findings in this study suggest that tree-ring analysis of *Pinus sylvestris* can be a useful approach for assessing the effects of drainage in peatlands. Analyses of these tree cores could potentially be used to identify drainage periods and estimate the magnitude of historical hydrological change, as was partially used in a study with *C. Thyoides* (Atkinson et al., 2025).

Further studies including a larger number of areas, longer time series after rewetting, and water table levels from before and after rewetting taken near each tree, could improve the understanding of how rewetting affects the growth in pine trees. Future research is also needed to investigate whether cores from pine trees could actually act as reliable indicators of the effect of drainage and rewetting, and if studying tree cores could be a future method for gaining knowledge about historical drainage and its effects. In addition, microscopic analyses of xylem cells could improve the understanding of cellular adaptations to changes in water table levels.

Conclusion

Overall, the study detected no statistically significant effect of rewetting on pine tree growth. However, an observed negative trend indicates that rewetting has a negative impact on tree growth. The results also indicate that increasing distance from the ditch may be associated with reduced growth. The lack of strong effects may be explained by site-specific differences in drainage intensity, rewetting effectiveness, and variable water table changes. Future studies are needed to better understand the effects of rewetting on pine trees.

Acknowledgements

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Data

Map of the sites: https://www.google.com/maps/d/u/0/edit?mid=1KRXLy3ZAfFDRPb5CTv-AUnD21JhMXn8&usp=sharing
Raw data & R-code can be found here: https://data.mendeley.com/datasets/yv9yrgt6yw/1
DOI:10.17632/yv9yrgt6yw.1

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